PARTIAL DIFFERENTIAL EQUATIONS

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Two – dimensional wave equation

- Fundamental PDE modeling motion of an elastic membrane, and gives the displacement distribution, $u(x, y, t)$ of a point (x, y) of the membrane from rest $\mathrm{u} = 0$ at time t

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$u = 0, \qquad for\ all\ t \geq 0$$

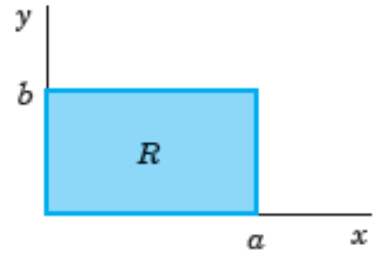
$$u(x, y, 0) = f(x, y)$$

$$u_t(x, y, 0) = g(x, y)$$

Two – Dimensional Wave Equation

2 – D wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$



Rectangular membrane

$$u = 0, \qquad for\ all\ t \geq 0$$

$$u(x, y, 0) = f(x, y)$$

$$u_t(x, y, 0) = g(x, y)$$

Solution Procedure

- Three ODEs from the wave equation

$$u(x, y, t) = \mathrm{F}(x, y)\mathrm{G}(t)$$

- Satisfying the boundary condition

$$u_{mn}(x,y,t) = (\mathrm{B}_{mn}\cos\lambda_{mn}t + \mathrm{B}^{*}_{mn}\sin\lambda_{mn}t +)\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}, \qquad m = 1,2,\ldots \ \&\ n = 1,2,\ldots$$

- These functions are called the eigenfunctions, or characteristic functions, and the values λ_{mn} are called the eigenvalues, or characteristic values.

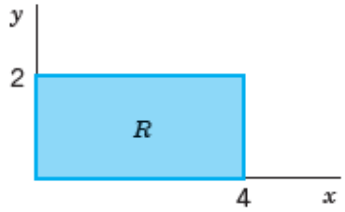
- Solution of the model: Double Fourier Series

$$u(x,y,t) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} u_{mn}(x,y,t) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} (\mathrm{B}_{mn}\cos\lambda_{mn}t + \mathrm{B}^{*}_{mn}\sin\lambda_{mn}t +)\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

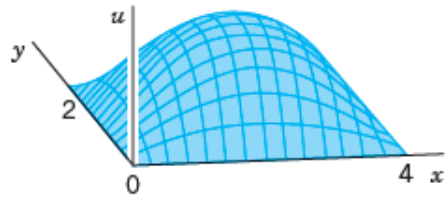
Sample Problem

Find the vibrations of rectangular membrane of sides a = 4 ft. and b = 2 ft. if the tension is 12.5 lb/ft., the density is 2.5 slugs/ft^2, the initial velocity is 0, and the initial displacement is

$$f(x,y) = 0.1(4x - x^2)(2y - y^2)$$



Membrane



Initial Displacement

Two – dimensional wave equation in polar coordinates

- Fundamental PDE modeling motion of an elastic circular membrane

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}\right)$$

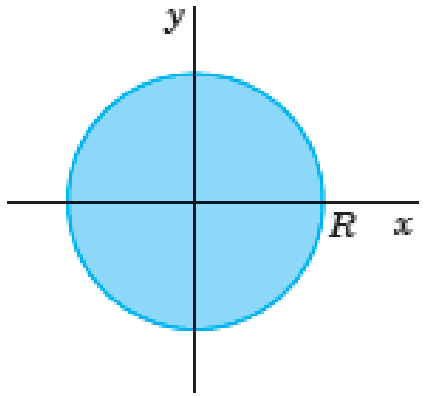
- Consider membrane of radius R

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r}\right)$$

$$u(R,t) = 0, \qquad for\ all\ t \geq 0$$

$$u(r,0) = f(r)$$

$$u_t(r,0) = g(r)$$



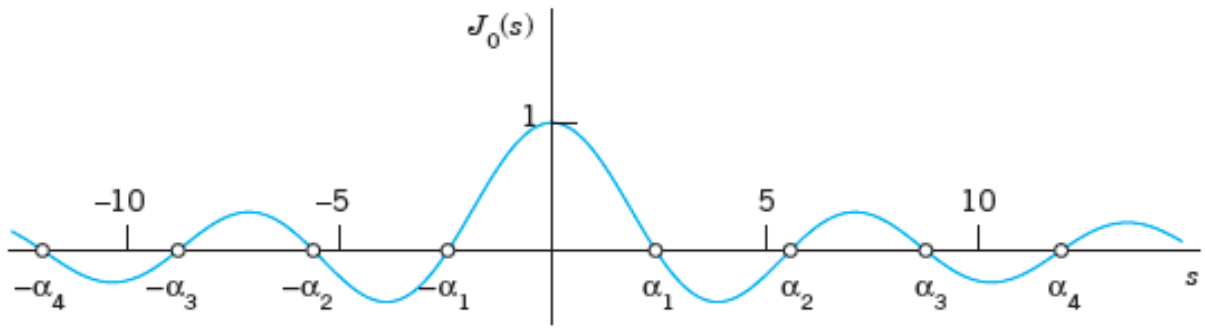
Circular membrane

Solution Procedure

- Two ODEs From the wave equation: Bessel's Equation

$$u(r,t) = \mathrm{W}(r)\mathrm{G}(t)$$

- Satisfying the boundary condition


$J_0(s)$
1
-10
-5
5
10
$-\alpha_4$
$-\alpha_3$
$-\alpha_2$
$-\alpha_1$
α_1
α_2
α_3
α_4
s

- Solution of the entire problem

$$u(r,t) = \sum_{m=1}^{\infty} u_m(r,t) = \sum_{m=1}^{\infty} (\mathrm{A}_m \cos \lambda_m t + \mathrm{B}_m \sin \lambda_m t)\mathrm{J}_o\left(\frac{\alpha_m}{R} r\right)$$

Laplace's equation

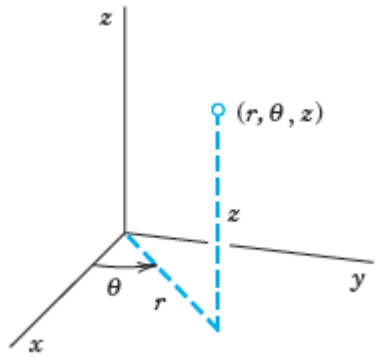
- Laplace's equation occurs mainly in gravitation, electrostatics, steady – state heat transfer and fluid flow. In Cartesian coordinates, the Laplace's equation is:

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}=0$$

$$\nabla^2 u=0$$

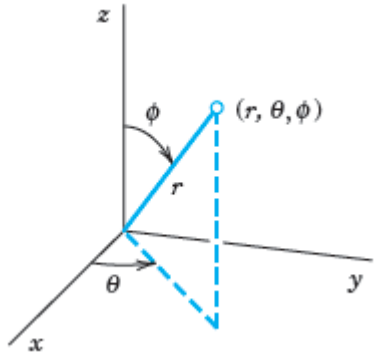
- Laplace's equation in cylindrical coordinates

$$\frac{\partial^2 u}{\partial r^2}+\frac{1}{r}\frac{\partial u}{\partial r}+\frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}+\frac{\partial^2 u}{\partial z^2}=0$$



- Laplace's equation in spherical coordinates

$$\frac{1}{r^2}\left[\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right)+\frac{1}{\sin\phi}\frac{\partial}{\partial \phi}\left(\sin\phi\frac{\partial u}{\partial \phi}\right)+\frac{1}{sin^2\phi}\frac{\partial^2 u}{\partial \theta^2}\right]=0$$



- Boundary value problem in spherical coordinate

$$\frac{1}{r^2}\left[\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right)+\frac{1}{\sin\phi}\frac{\partial}{\partial\phi}\left(\sin\phi\frac{\partial u}{\partial\phi}\right)\right]=0$$

$$u(\mathrm{R},\phi)=f(\phi)$$

$$\lim_{r\to\infty}u(r,\phi)=0$$